

MECH 330 – APPLIED THERMODYNAMICS II

Course Syllabus – Fall 2026

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This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.
See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/vpcei/support/overview>.

COURSE INSTRUCTOR

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Engineering

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Office hours: 2-3 PM Thursdays



MECH 330 (F 3-0-0.5 3.5)

COURSE DESCRIPTION

A continuation of MECH 230 with selected topics such as gas and vapour power cycles, refrigeration, mixtures of gases and vapours, combustion, and available energy.

(0/0/0/42/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

PRE-REQUISITE KNOWLEDGE

This course builds on knowledge from the successful completion of prerequisite courses, some review will be covered in first weeks of course.

Prerequisites: MECH 230 (Applied Thermodynamics I) or ENPH 274 (PHYS 274)

MODE OF INSTRUCTION

The delivery mode of this course is in-person.

COURSE LOCATION AND TIME

Lectures:

- Monday 9:30 AM – 10:30 AM (Biosci 1101)
- Thursday 8:30 AM – 9:30 AM (Biosci 1101)
- Friday 1:30 PM – 2:30 PM (Dunning AUD)

Tutorials:

- Monday 8:30 AM – 9:30 AM (Dunning 12) – Section 002
- Wednesday 8:30 AM – 9:30 AM (Dunning 11) – Section 003
- Wednesday 4:30 PM – 5:30 PM (Dunning 11) – Section 004
- Friday 9:30 AM – 10:30 AM (McLaughlin 315) – Section 005

COURSE LEARNING OUTCOMES (CLO)

By the end of this course, students should be able to:

CLO	DESCRIPTION	INDICATOR
CLO 1	Calculate energy efficiency changes using re-heating, regeneration, use of open and closed feedwater heaters, and deaerators in vapour power systems (Vapour power)	KB-Engineering Science (Thermodynamics), PA – Identify, PA – Formulate, PA – Solve

CLO	DESCRIPTION	INDICATOR
CLO 2	Calculate energy efficiency changes associated with regenerative heating through use of heat exchangers, reheat and intercooling in gas power systems or Brayton Cycles (Gas power)	KB-Engineering Science (Thermodynamics), PA – Identify, PA – Formulate, PA – Solve
CLO 3	Use exergy analysis to calculate energy availability and effectiveness (Exergy)	KB-Engineering Science (Thermodynamics), PA – Identify, PA – Formulate, PA – Solve
CLO 4	Calculate the coefficients of performances of vapour compression and Brayton refrigeration cycles that use different working fluids with multistage compression and intercooling in refrigeration and heat pump systems (Refrigeration)	KB-Engineering Science (Thermodynamics), PA – Identify, PA – Formulate, PA – Solve
CLO 5	Calculate thermodynamic properties of ideal gas mixtures, including gases that contain water vapour and apply the Psychrometric chart (Psychrometrics)	KB-Engineering Science (Thermodynamics), ET – Apply, ET - Limitations
CLO 6	Calculate the energy release from combustible mixtures, including the lower and higher heating values (Combustion)	KB-Engineering Science (Thermodynamics), PA – Identify, PA – Formulate, PA – Solve

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date (subject to change per onQ)	Weight	Alignment with CLOs
Traditional Assignments	Mondays by 11:59PM	5% (1 dropped)	All
A1	Sept. 21	1%	4
A2	Oct. 5	1%	3,4
A3	Oct. 20 (Tuesday)	1%	1
A4	Nov. 2	1%	1,2

Assessment Tool	Due Date (subject to change per onQ)	Weight	Alignment with CLOs
A5	Nov. 16	1%	2,5
A6	Dec. 7	1%	6
Jupyter Assignments	Mondays by 11:59PM	5% (1 dropped)	1,2,4
J1	Sept. 28	2.5%	4
J2	Oct. 26	2.5%	1
J3	Nov. 9	2.5%	2
Mid-term 1	Thursday, October 22, 6 PM	25%	3,4
Mid-term 2	Wednesday, November 25, 6 PM	25%	1,2,5
Final	TBD	40%	All
		100%	

ASSESSMENT DESCRIPTIONS

Traditional Assignments (5%)

There are 6 traditional assignments due approximately every other week, submitted electronically on onQ. The assignments will not be marked in detail, instead learners are encouraged to consult homework solutions posted on onQ for feedback. They will be reviewed for **original content** and graded as 1 – strong effort, 0.5 – moderate effort, and 0 – minimal effort or missing. The top 5 assignments will be used for the overall assignment mark (one will be dropped) to accommodate for extenuating circumstances. Students should aim to submit all assignments. **Each student is responsible for completing their own traditional assignments.** Late assignments will not be accepted, except with documented accommodation or permission from the Faculty office.

Jupyter Assignments (5%)

In addition to the traditional assignments, there will be 3 'Jupyter Assignments' which are assignments completed in a python-based Jupyter notebook. These assignments will tackle in-depth problems using the thermodynamic computational library Cantera, also submitted electronically on onQ. These assignments are designed to introduce learners to modern tools to model energy systems. A Jupyter notebook skeleton with questions and starter scripts will be provided for every assignment. Students may choose to work with one partner and submit a single assignment for both group members. Students may choose their own

partner at the beginning of the term on onQ. This partner is fixed for the duration of the term. Jupyter Assignments will be graded similarly to Traditional Assignments for **original content** and graded as 2.5 – excellent effort, 2 – strong effort, 1 – low effort, and 0 – negligible effort or missing.. The top 2 Jupyter assignments will be used for the overall assignment mark to accommodate for extenuating circumstances. Late assignments will not be accepted, except with documented accommodation or permission from the Faculty office.

Mid-terms (50%)

Two mid-terms are planned (25% weight each), tentatively scheduled for Thursday October 22 and Wednesday November 25, both at 6 PM. The mid-term is closed book, but a formula sheet will be provided. If needed, steam tables or other necessary properties will be provided. You should not schedule vacations, travel, etc. during the mid-term period, as these are not valid reasons for a make-up exam or mark redistribution.

Final (40%)

The final is closed book, but a formula sheet will be provided. If needed, steam tables or other necessary properties will be provided. Students must write their exam on the day and time scheduled by the University. You should not schedule vacations, travel, etc. during the exam period. The [Term and Session Dates](#) will indicate the final exam period session dates in each term.

GRADING

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established [Grade Point Index](#).

Feedback on Assessments

The teaching team will provide feedback on exams (for Assignments, see above). You can expect feedback on your assessments within seven days of the exam date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

Required Textbook

Fundamentals of Engineering Thermodynamics 9th Ed. by Moran, Shapiro, Boettner, and Bailey, Wiley. A paper copy or e-book is acceptable. The steam tables in the back of the book are needed to complete traditional assignments, and students should be comfortable working with these tables because they will also be used in exams (a copy of the tables will be provided in exams).

Cost: \$153.95 (print copy), \$134.95 (e-book), \$60.00 (e-book rental). An earlier edition is ok, although there may be some mismatch between assigned reading.

Other Material

All other course material is accessible via OnQ.

Required Calculator

A Casio 991 is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one semester spanning 12 weeks. Learners can expect to invest on average 7-9 hours per week in this course. Learners who adhere to a pre-determined study schedule are more likely to successfully complete the course.

WEEKLY COURSE LEARNING OUTCOMES

Note that the following is approximate timing only, and the content will be adjusted based on course progression. Weekly readings may deviate from below and will be assigned during the course of the term.

Week	Learning Outcomes	Assessment
1	Introduction Reading: Review Chapters 3-6; Supplemental handout entropy By the end of this week, learners will be able to: • Understand the motivation for studying energy systems • Recall MECH 230 (or equivalent) material	None
2	Vapor-compression refrigeration Reading: 10.1-10.4 By the end of this week, learners will be able to: Analyze thermodynamic models of vapour compression systems and their modifications, including T-s diagrams, property evaluation, applying mass, energy, entropy balances, COP.	None
3	Advanced refrigeration and heat pumps Reading: 10.5-10.6 By the end of this week, learners will be able to: Analyze advanced vapor compression refrigeration and heat pump cycles and absorption cycles.	A1

Week	Learning Outcomes	Assessment
4	Available energy (exergy) Reading: 7.1-7.6, 9.9 By the end of this week, learners will be able to: • Perform exergy balances to closed systems and control volumes • Define and calculate exergetic efficiencies	J1
5	Vapor power systems Reading: 8.1-8.2 By the end of this week, learners will be able to: • Demonstrate basic understanding of vapour power systems and modifications including T-s diagrams, evaluating property data at principal states, apply mass, energy, and entropy balances, determine cycle performance, thermal efficiencies, net power output and mass flow rates	A2, Midterm 1
6	Vapor power systems continued Reading: 8.3-8.4 By the end of this week, learners will be able to: • Understand and analyze advanced vapor power system modifications such as reheat, superheat, and regeneration	A3

Week	Learning Outcomes	Assessment
7	Gas power systems Reading: 9.5-9.6 By the end of this week, learners will be able to: Perform air standard analyses of gas turbine power cycles based on Brayton cycle including: T-s diagrams, evaluating property data at principal states; applying mass, energy, and entropy balances; determine net power output, thermal efficiency, back work ratios, and compressor pressure ratio	J2
8	Gas power systems continued Reading: 9.7, 9.8, 9.11, 10.7 By the end of this week, learners will be able to: Perform more advanced analysis of gas turbine systems including regenerative gas turbines and gas turbines for aircraft propulsion. Analyze thermodynamic models of gas-powered refrigeration.	A4
9	Psychrometrics Reading: 12.1-12.7 By the end of this week, learners will be able to: - Ideal gas mixtures using molar and mass fractions - Dalton model to calculate state properties - Apply mass, momentum and energy involving gas mixtures - Humidity ratio, relative humidity, mixture enthalpy, dew point temperatures - Plot processes on a psychrometric chart	J3

Week	Learning Outcomes	Assessment
10	Combustion Reading: 13.1-13.3 By the end of this week, learners will be able to: • Demonstrate understanding of key concepts of complete combustion, theoretical air, enthalpy of formation • Balance stoichiometric combustion equations	A5
11	Combustion continued Reading: Lecture notes By the end of this week, learners will be able to: • Calculate adiabatic flame temperature • Understand combustion kinetic pathways, chain reactions, and pollutant formation	Midterm 2
12	Advanced technologies and the energy landscape Reading: Lecture notes/slides By the end of this week, learners will be able to: • Critically assess new technologies and their potential for impact • Quantify the environmental impact of leading energy technologies	A6

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

CONFIDENTIAL MATTERS

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the [Smith Engineering website](#). Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

For Traditional Assignments and Jupyter Assignments, 1 assignment is automatically dropped for each category for extenuating circumstances. If a midterm is missed in the case of an approved consideration request, the marks are redistributed equally to the remaining midterm and the final (10% to each exam).

LATE SUBMISSION POLICY

In the event of extenuating circumstances, you must follow the policies for requesting an academic consideration (as described above) or engage the appropriate Academic Accommodation process (see Academic Accommodation section below). In the absence of an approved consideration request, or accommodation in VENTUS, late assignments will not be accepted.

REQUESTING ACADEMIC CONSIDERATION RELATED TO GROUP WORK

The group assignments (Jupyter Assignments) should take students 2 days to complete, but to suit the needs of all learners, you have been given 7 days to complete the group assignment. As such, no further accommodations or extensions will be applied for the group assignment.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

NETIQUETTE

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.

<https://www.queensu.ca/nonacademicmisconduct/policies>

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's **express**

consent. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the [Professional Engineers Ontario Code of Ethics](#). Information on policies concerning academic integrity is available in the [Queen's University Code of Conduct](#), in the [Senate Academic Integrity Policy Statement](#), on the [Smith Engineering website](#), and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are permitted in this course under the following circumstances:

1. GenAI tools are permissible only if explicitly noted in the activity and/or assignment instructions.

2. You may use GenAI tools for the following uses: general help with concepts, help with general programming tasks (e.g., plotting, for loops, making lists, etc.)
3. You are restricted from using GenAI tools for: fully answer assignment questions, perform thermodynamic calculations. I will note, in addition, that trials of ChatGPT 5 (Fall 2025 version) answering Jupyter Assignment questions yielded generally reasonable code but with some serious errors which may be difficult to find.
4. If any GenAI tools have been used, you must disclose the use and purpose and include proper citation of the generated materials. For information on how to cite the use of GenAI tools, please visit [Queen's University Library](#).

INVALID EXAMS

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the [Library](#), [Student Academic Success Services \(Learning Strategies and Writing Centre\)](#), and [Career Services](#). Learners are encouraged to visit the Smith Engineering [Current Students](#) web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

ABSENCES - ACADEMIC CONSIDERATIONS

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

ABSENCES – REQUESTS FOR EXCUSED ABSENCES FOR SIGNIFICANT EVENTS (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date. For additional information and to submit a REASE application, please review the information on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

ACADEMIC ACCOMMODATIONS

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see [Smith Engineering's Academic Accommodation website](#).

RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the Religious Observance and Academics Activities form available on [Smith Engineering's Academic Accommodation webpage](#) within a week of receiving their course syllabus (each term).

- Alternative assignments are considered a “reasonable accommodation” under the Ontario Human Rights Code
- Please refer to the [Queen's Multi-faith Calendar](#) for all of this year's faith dates. Requests for Religious Accommodation will be verified based on the dates in this calendar. If you are requesting dates not listed in this calendar, you may be asked for additional documentation
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca or Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Further information can be found on the [Smith Engineering's Academic Accommodation webpage](#)

OTHER HUMAN-RIGHTS BASED ACCOMMODATIONS

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Smith Engineering Academic Accommodation and Consideration Advisor at <mailto:engineering.aac@queensu.ca> or to call 613-533-6000 x 78139.

ACCOMMODATIONS FOR DISABILITIES

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with

disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the [QSAS website](#) to learn more about academic accommodation.

To start the registration process with QSAS, click the **“Access Ventus”** button found on the [Ventus student portal](#).

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:

<https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students>

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Academic Accommodation and Consideration Advisor at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

STUDENT WELLBEING

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, [EngWell](#) is here to help.

EngWell provides confidential support through our [Wellness Advisor](#), [Embedded Counsellors](#), [mental health programming](#), and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, [we encourage you to reach out early](#).

EDUCATIONAL TECHNOLOGY

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
MS-Teams	https://support.microsoft.com/en-us/office/accessibility-support-for-microsoft-teams-d12ee53f-d15f-445e-be8d-f0ba2c5ee68f

Zoom	https://zoom.us/accessibility
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If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

TECHNICAL SUPPORT

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact [Technical Support](#).